Supplementary Information: Experiment-Driven Bayesian Optimization of the Impact Response of Multi-Material 3D-Printed Tensegrity-Inspired Structures

Draft note (remove before submission): as in the main text, any value printed in this violet color is a synthetic placeholder rather than a measurement. As of 2026-09-15 this Supplementary Information contains no violet values. The former placeholder batch-2 print key was replaced by the logged campaign record, which now covers batches 2 to 4.

This Supplementary Information collects the detailed design-, fabrication-, test-, and analysis-development records supporting the main manuscript, together with the tables and figures moved out of the main text to meet the journal's length limit (the main text cites them by their S-numbers). Unlike the main text, the SI references the project's development history directly, including specific pull requests (PRs) and issues in the project repository (https://github.com/vertical-cloud-lab/tensegrity-optimization), whose contents are listed in the Data Availability Statement of the main text. Citations use the reference list at the end of this document, which is numbered independently of the main text.

S1 Positioning Against Prior Work

Table S1 positions the present campaign against selected prior efforts spanning the three elements it combines: impact-tested printed tensegrity-inspired structures, rigid-flexible additive fabrication, and Bayesian optimization (BO) of architected materials. The main text (Section 2) discusses each of these studies. None of them combines a multi-material fused-deposition-modeling (FDM) tensegrity-inspired architecture with an experiment-driven optimization loop.

S2 Printability Screens and As-Built Resolution

The constant-mass projection of the main text (Section 3.1) scales every candidate design before printing. This section records the two printability screens applied to the projected designs and what the print record shows about them, and the as-built scatter in mass.

Printability screens. We also checked each projected design against two printability limits. Both were recorded as warnings, and neither was used to reject or change a design. The campaign code evaluates both on the as-printed geometry of every recommended design (`bo/t3_prism_bo_campaign.py` on the campaign branch, PR #102). It flags a design in the batch tables as "envelope > 250 cm^3" or "cable < 3.0 mm". The first compares $\pi R^2 H$ with $250\,\mathrm{cm}^3$, where R is the radius of the circle through the joint centers and H is the height between joint

Table S1: Positioning of this work against selected prior efforts spanning the elements combined here. The present campaign uses physical drop measurements of the transmissibility (t_{180}) and the mass-weighted rebound index (I_{reb}, Eq. (1) of the main text, whose first-power form E_{reb} recommended batches 2 to 4) as its BO responses. It does not optimize specific energy absorption or crush energy.

Study	Architecture	Fabrication	Optimization	Response optimized	Ground truth
Pajunen et al. 2019 [1]	Truncated-tetrahedral tensegrity-inspired	Single-material 3D print	None (single condition)	n/a	Experiment (drop impact)
Intrigila et al. 2022 [2]	Bistable tensegrity-like lattice unit cell	Single-material 3D print	None	n/a	Experiment (quasi-static compr.)
Mo et al. 2023 [3]	Architected lattices	n/a (simulation)	Multifidelity BO	Simulated response metric	Simulation (+ sparse high-fid.)
Almeida 2025 [4]; Davami 2025 [5]	Tensegrity-inspired structures	Additive manufacturing	None reported	n/a	Experiment (high-rate/dynamic)
Wang et al. 2026 [6]	Tensegrity-inspired rigid-flexible metamat.	Integrated dual-material	None reported	n/a	Physical validation
This work	Tensegrity-inspired T_3-prism cell (PLA/TPU)	Multi-material FDM (PLA + TPU)	Experiment-driven BO (SAASBO/qNEHVI)	t_{180}; provisional I_{reb}	Physical experiment

centers. The 250 cm^3 value is a placeholder size limit carried over from an earlier simulation study. It is not the printer's build volume, and because it is measured to the joint centers it undercounts the printed part. Seven printed specimens (five designs) exceeded it. The second limit is a 3.0 mm minimum on the printed TPU cable diameter, adopted after the thin, unsupported top cables of early PETG test prints failed. Because the projection scales the whole cell, a design at the 3.0 mm lower bound of the cable variable can print thinner. Nine specimens printed with cables of 2.51 to 2.78 mm, including all four batch-4 specimens that reduced the peak shock (main text, Section 4.2). With hand-painted tendon supports (Section S4), all nine printed. One had stringing, one had a slight tendon indent, and seven had no logged defect. The most common tendon defect was a TPU strand that did not fuse to the tendon and came away from it. It was logged on five specimens, all with cables of 3.28 to 4.23 mm. The 3.0 mm value is therefore a conservative rule for this workflow. Our prints do not show that thinner cables cause more defects.

As-built resolution. The nozzle diameter, the slicer line width, and support interaction discretize the as-built cross-sections, so the effective search granularity is coarser than the nominal continuous box. The 3.0 mm lower bound on the TPU cable diameter is likewise a limit of this workflow. It is the printability minimum described above, set by printing unsupported cables and by the slicer's auto-support detection on the H2D printer used in this study. Two sets of repeated prints bound the fabrication scatter in mass. One seed design printed three times has a mass standard deviation of 0.457 g, the value used in the observation noise model (main text, Section 3.4). The batch-3 plate, also printed three times, averaged 19.59, 19.86, and 20.07 g per print, so its second and third prints ran 0.27 g and 0.48 g heavier specimen for specimen. That is a per-print flow offset rather than random scatter.

S3 Joint-Design Study

We developed the strut-to-cable junction through an OpenSCAD study of five candidate joints (Designs A to E: anchor-bulb, dovetail, TPU-sleeve overmold, eyelet-loop, and TPU-rebar), followed by a sixth, captive-core design (Design F). Every campaign specimen uses Design F, a captive TPU core inside a PLA shell. At each strut end the tendons meet in a TPU core that is too large to

pass back out through the shell's tendon bores. The joint therefore holds mechanically even though PLA and TPU bond weakly, and the shell keeps the soft tendon away from layer-line failure at the transition. The campaign CAD (`cad/t3-prism/t3-prism.scad` on the campaign branch) builds this joint with `use_captive_core = true`.

Full results, renders, and the Phase-3/Phase-4 design reviews are documented in PR #39 (joint design) and PR #35 (CAD parameterization and print history).

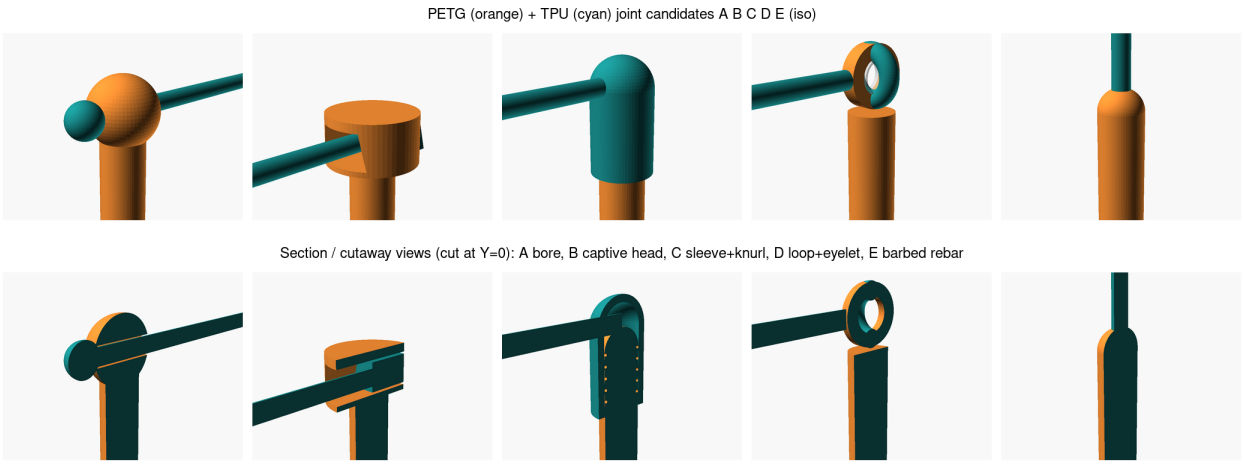


Figure S1: Five candidate strut-to-cable joint geometries from the OpenSCAD joint study (left to right: A anchor-bulb, B dovetail, C TPU-sleeve overmold, D eyelet-loop, E TPU-rebar; top row isometric, bottom row section views). The study renders use the polyethylene terephthalate glycol (PETG, orange) and TPU (cyan) material pair under which the joint family was first developed. The campaign specimens use none of these five. They use the later captive-core joint (Design F), a TPU core inside a PLA shell, with the PLA/TPU pair of the main text.

S4 Support Strategy for Soft Members

Near-vertical TPU tendons are otherwise unsupported during printing. For the printed campaign batches, we disabled automatic support generation and painted PLA supports manually in Bambu Studio along the edge of each tendon. The support type is "Tree (manual)", which grows tree supports only from the painted regions (Table S2). We tightened the support-to-object XY clearance from 0.35 mm to 0.2 mm during tuning on the S0 reference prism (issue #98). Support removal is the leading source of the cosmetic defects that we log per specimen in the print key (Section S6). They are surface stringing and, on two seed specimens, a TPU strand that did not fuse to a bottom tendon and came away from it.

In parallel, we built an automated alternative in software and archived it, but no campaign specimen used it. Every campaign specimen was printed with the manually painted supports described above. The alternative combines a tensegrity-specific Bambu Studio recipe (support threshold angle lowered from 40° to 10°, support material matched to the rigid extruder) with narrowing support pillars placed by ray-casting against the underside of the printable mesh. The narrowing-pillar workflow is documented in PR #66, and the H2D multi-part material assignment fix in PR #64.

S5 Print Parameters and Process Control

Table S2 lists the multi-material print parameters, transcribed from the archived as-printed Bambu Studio project. The joint geometry is described in Section S3 and the support strategy in Section S4.

Moisture and nozzle management. TPU is hygroscopic, so we fed the soft filament from a heated dry box held near or below 8 % relative humidity, logged per print. Midway through the seed batch the TPU extruder developed a progressive drop in flow (a partial clog, compounded by a stray PLA fragment found inside the extruder). We resolved it by performing a cold pull and moving the TPU to a dedicated high-flow 0.6 mm nozzle, after which print quality was restored. We identify the affected and post-fix specimens in the print key so that any nozzle-related covariate can be checked against the measured objectives.

Table S2: Multi-material print parameters on the Bambu Lab H2D, transcribed from the archived as-printed Bambu Studio project (`t3-prism-bo-batch.H2D-MM-PLAstruts-TPUcables.as-printed.3mf`, Bambu Studio 2.7.1). Filaments are Bambu Lab PLA Basic (struts, support material) and Bambu Lab TPU 85A (tendons). The print profile is "0.30 mm Standard @BBL H2D 0.6 nozzle."

Parameter	PLA (struts)	TPU 85A (tendons)
Nozzle	0.6 mm	0.4 then 0.6 mm HF[a]
Nozzle temperature (°C)	220[b]	240[b]
Max volumetric speed (mm^3/s)	30[b]	2.5[b]
Bed temperature, profile (°C)	55	35
Layer height / line width (mm)	0.30 / 0.62	
Infill	15% grid, 2 walls	near-solid
Outer / inner wall speed (mm/s)	200 / 300 (profile)	
Filament drying	n/a	dry box, near or below 8% RH
Build-plate type	Textured PEI	
Build orientation	Upright (main-text print timelapse figure)	
Supports	Tree (manual), PLA, only where painted[c]	

[a]The TPU side ran a standard 0.4 mm nozzle until the mid-batch clog, then a dedicated high-flow (HF) 0.6 mm nozzle. Affected specimens are flagged in the print key. [b]Seed and batch-2 plates, as recorded in their archived slicer projects (first TPU layer at 235 °C). Batch 3 printed PLA at 217 °C and 20.5 mm^3/s and TPU at 244 °C and 3.6 mm^3/s. Batch 4 used 226 °C and 29.5 mm^3/s for PLA and 236 °C and 2.6 mm^3/s for TPU (main text, batch lines of the model-recommended batch-results table). The wall speeds ask for more flow than either limit allows, so the volumetric limit sets the extrusion rate of both materials. [c]Bambu Studio's "Tree (manual)" support type grows tree supports only from regions painted with the support tool. We painted them along the edge of each tendon (Section S4). The batch-3 and batch-4 slicer projects archived with the campaign code record supports as off, so the supports of those batches rest on the print log.

S6 As-Printed Campaign Record (Print Key)

Table S3 is the print key for the seed campaign: every specimen's print identifier, its design, its weighed mass, and its logged defects. The authoritative machine-readable key (`bo/t3-prism-bo-batch-print-key.csv`) and the as-printed slicer project (one specimen per named build plate) live at PR #102. Two entries deserve care. We printed design 08 three times. The plate label in the archived slicer project and a contemporaneous issue comment disagree on which specimen is the official one (*dea4ls* versus *bag26v*), and pending confirmation we use *bag26v*

in the analysis. The triplicate prints of design 08 also supply the only print-to-print repeatability number for a seed design, a mass standard deviation of 0.457 g. Three prints of a single geometry do not give a general process repeatability estimate. Separately, one tested specimen (*amdjwm*) has no confirmed design mapping and is excluded from the surrogate fit until identified.

Table S3: Print key for the seed campaign (nine Sobol designs plus the S0 reference). Masses weighed on a 0.01 g scale. TPU fed from a heated dry box near or below 8% relative humidity where logged. The first five specimens (printed before the TPU nozzle fix) used the standard 0.4 mm TPU nozzle, and the rest used the high-flow 0.6 mm nozzle.

Design	Specimen	Mass (g)	Logged defects
Spec 00	9hhbkp	21.62	none
Spec 01	6lhxfy	18.50	slight strut misprint; divot on one top tendon
Spec 02	autv5r	22.04	minor tendon misalignment near top
Spec 03	ebdna8	19.61	two strands separated from bottom tendons (support removal)
Spec 04	nvxsrv	20.66	misalignment ring after printer pause
Spec 05	6nheas	21.73	two strands separated from bottom tendons (support-related)
Spec 06	1zm8rv	18.50	stringing along diagonal tendons
Spec 07	ajhby6	20.53	filament line pulled from bottom struts
Spec 08	dea4ls	22.29	stringing; residual PLA on tendon
Spec 08	bag26v	21.42	some stringing on tendon
Spec 08	ghmj4y	22.10	inconsistent tendon cross-section near top
S0 ref.	bpx68c	20.23	none major

Table S4 is the corresponding logged key for the three model-recommended batches. Every batch-2 and batch-3 specimen printed at about 11% dry-box relative humidity and every batch-4 specimen at about 7%. The batch-3 plate was printed three times. The second-print specimens (2dran1 to 2dran9) weighed 19.48 to 20.32 g, on average 0.27 g heavier than the matching first-print specimens, and were all logged defect-free. The third-print specimens (dran31 to dran39, printed 2026-09-28) weighed 19.68 to 20.56 g (mean 20.07 g), and the print log notes no defects. The lab record confirms each 2dran$N$ and dran3N as a re-print of drran$N$ at the same Ax trial, so the three prints pair specimen for specimen (Table S6). The nine second-print specimens entered the surrogate fit as distinct completed trials at their own weighed masses and per-session standard errors. The third print was drop-tested on 2026-09-28, 2026-09-29, and 2026-10-02 (20 drops each, all captures clean) and has not yet entered the fit. Its records are on PR #86. The authoritative machine-readable keys, with plate positions and the mapping evidence per row, are the `t3-prism-bo-round1/round3/round4-print-key.csv` files archived with the manuscript data.

S7 Drop-Tower Protocol and Rig Characterization

The campaign fixture is a Lansmont Model 23 shock test system with a Test Partner 4 data-acquisition console. This section summarizes the development history behind the protocol in the main text.

Protocol. The drop height is 60 in, with the specimen upright on the carriage plate. We discard the first 2 drops of each session as seating warm-up. Session length differs by batch. Seed sessions recorded 101 drops per specimen (103 for autv5r), batch-2 sessions 21 or 22, and batch-3 and batch-4 sessions 20. After the warm-up discard, 99 (101), 19 or 20, and 18 stabilized drops per

Table S4: Print key for the model-recommended batches, first prints only. The text summarizes the second and third prints of the batch-3 designs. Masses weighed on a 0.01 g scale.

Batch	Trial	Specimen	Mass (g)	Logged defects
2	10	r2d2c9	17.91	none
2	11	r2d2c4	19.33	string on one top tendon; partial TPU detachment
2	12	r2d2c2	17.96	none
2	13	r2d2c1	19.38	detached TPU string, two top tendons; odd foot support
2	14	r2d2c6	19.11	none
2	15	r2d2c7	18.51	none
2	16	r2d2c3	18.63	none
2	17	r2d2c8	19.98	full-span strings, two bottom tendons
2	18	r2d2c5	23.47	none
3	28	drran5	19.88	tiny bubbles, diagonal tendons
3	29	drran4	19.92	none
3	30	drran9	19.34	none
3	31	drran3	19.92	tiny bubbles, diagonal tendons
3	32	drran7	20.06	tiny bubbles, diagonal tendons
3	33	drran2	19.33	none
3	34	drran6	19.24	tiny bubbles, diagonal tendons
3	35	drran8	19.32	none
3	36	drran1	19.31	none
4	37	corny7	19.62	none
4	38	corny2	20.21	none
4	39	corny8	19.64	none
4	40	corny5	20.58	none
4	41	corny6	19.45	none
4	42	corny1	20.20	TPU mixed into one PLA strut layer
4	43	corny9	19.46	none
4	44	corny4	20.56	small pores, diagonal tendons
4	45	corny3	20.15	none

specimen enter the means. The per-specimen summaries archived with the data count every valid capture, including the 2 warm-up drops (column `n_valid`), and the raw series tables of the seed sessions hold 101 events each (103 for autv5r). Section 3.3 of the main text justifies the shortening. The statistical minimum from the pre-campaign variance study is five stabilized drops, and a truncation replay of the seed sessions preserves the full-session transmissibility ranking at 20 drops. The three prints of batch 3 also showed that the error budget is dominated by seating and print variation rather than session length. We record three high-speed video clips (959 fps) per specimen. From batch 4 on, we film them at drops 1, 10, and 20 so that clips pair deterministically with captures. The camera cannot resolve the millisecond-scale impact pulse (one to two frames), so the accelerometer channels produce the objectives and the video documents the post-impact rebound and recovery. Synchronized laser-Doppler vibrometry is reserved for follow-up characterization of selected designs. Testing takes roughly 42 s per drop, so about 90 minutes per seed specimen and 15 per later-batch specimen. Capture settings: 1.25 MHz sample rate, 100 ms record, 2 ms pre-trigger, 150 G trigger on the base channel, AC-coupled ICP conditioning. After a settings-loss event, we

re-verified the console settings digit-for-digit against an archived screenshot and confirmed data continuity to within about 1%.

Impact surface. A blinded 3×3 crossover experiment over three candidate polyurethane mat arrangements (single 0.25 in Shore 70 OO, single 0.5 in Shore 60 OO, stacked) selected the single 0.5 in Shore 60 OO sheet (issue #88, PR #86). Unlike the earlier felt stack, the mat does not measurably compact over 100-drop sessions. Sessions recorded as "5 felts" in early protocol documents were physically four felt sheets plus one cardboard layer.

Sensors and calibration. Base input: single-axis piezoelectric accelerometer on the carriage plate (1.059 mV/G, full scale 9443 G). Top response: Dytran 3133A4 triaxial accelerometer (X/Y/Z 0.690/0.667/0.734 mV/G) seated in a printed pocket at the top vertex, mounted per ISO 5348. We retired an earlier single-axis channel after range clipping. We cross-calibrated the two base-capable channels in place over 15 co-located drops. The ratio was 0.953 ± 0.001 at CFC-180 (per-drop 0.952 ± 0.004) and stable across drop heights (PR #74).

Filtering correction. We implemented the SAE J211/1 channel-frequency-class filter used for all reported numbers from the standard's published coefficients. An earlier in-house implementation passed the two-pass (forward-backward) corner frequency to a single-pass design, which narrows every channel class by roughly 20% (an effective CFC-144 where CFC-180 was intended). We identified the discrepancy during the analysis audit of issue #94 and corrected it before the campaign numbers were produced.

Processing pipeline. For each capture, we calibrate and CFC-filter every channel independently. We form the top-vertex magnitude $\|\mathbf{a}_{\text{top}}(t)\|$ from the three filtered triaxial components and search for the two peaks entering t_{180} within a short window (approximately 1.5 ms) of first contact, so the peaks need not be simultaneous. We estimate the baseline from the record tail, because at healthy tower speed the mat-contact foot begins before the 2 ms pre-trigger window. Δv is the integral of the baselined base channel over the impact pulse and doubles as the rig-health check. Drops are discarded per the warm-up rule. Table S5 collects the operational analysis parameters in one place. Values known from the protocol are stated numerically. The remaining thresholds are to be transcribed verbatim from the version-tagged campaign analysis code at its archival release, so that the table matches the code of record.

Rig health and repair history. We check rig health against the impact velocity change Δv recovered from the base channel. Over an uninterrupted 100-drop session the Δv coefficient of variation (CV, the standard deviation divided by the mean) was 0.54% with a flat drift slope. Within a session, Δv and the CFC-180 input typically ease down by 1 to 4% as the polyurethane mat warms, and recover fully after a rest of minutes. After a mat rest or re-seat, the first 5 to 10 drops carry the settling transient, which is why the warm-up discard exists. The arrest velocity has visibly stabilized as cumulative drop count grew, so we keep the Δv record. During commissioning, two guide-rod locking pins failed, and we discovered and recommissioned a defeated pin-park safety interlock. The pin-related friction produced a measurable Δv deficit that resolved after repair (issue #92). We have performed more than 4,400 instrumented drops across roughly 80 printed specimens over about four months of commissioning, characterization, and campaign sessions (tallied through issue #101 plus the batch records). The campaign batches themselves

Table S5: Operational analysis parameters for the drop-tower pipeline. "Code of record" marks values to be transcribed from the version-tagged analysis release.

Parameter	Value
Drop height	60 in (1.524 m)
Sample rate / record / pre-trigger	1.25 MHz / 100 ms / 2 ms
Trigger	150 G on the base channel
Reporting filter class	CFC-180 (SAE J211/1), two-pass corner corrected
Diagnostic filter class	CFC-1000
Peak-search window	≈1.5 ms from first contact
Baseline estimate	record tail
Warm-up exclusion	first 2 drops per session
Drops recorded per session	101 seed (103 autv5r); 21-22 batch 2; 20 batches 3-4
Stabilized drops per specimen	99 seed (101 autv5r); 19-20 batch 2; 18 batches 3-4
Rig-health check	Δv from baselined base channel
Ringdown fit gate	$r^2 \geq 0.85$ (exponential fit)
Second-contact threshold and window	code of record
Per-capture exclusion rules	code of record
Stabilized-mean convention	code of record
Filter initialization / boundary handling	code of record

contribute about 1,500 stabilized captures including the second print of batch 3, and its third print adds 162 more.

Ringdown diagnostics. We recorded ringdown descriptors per drop as print-health and pre-stress diagnostics [7], but they are not campaign objectives. They are the fitted natural frequency f_n and, where the exponential fit passes an $r^2 \geq 0.85$ gate, a damping ratio ζ. Of the 44 sessions with an accepted f_n, all but two rang at 260 to 470 Hz. The exceptions are r2d2c7 (682 Hz, batch 2) and corny8 (764 Hz, batch 4). In the seed batch, f_n was 368 Hz for 6lhxfy, 455 for amdjwm, 322 for 6nheas, 468 for bpx68c, 364 for 9hhbkp, 294 for nvxsrv, and 386 for autv5r, with no accepted estimate for ajhby6 or bag26v.

Seating sensitivity, the between-seat rule, and the seat gauge. The three prints of batch 3 (main text, Section 4.2) showed that a session can be internally stable and still misrepresent its design. The 1.251 session passed every within-session check, and the same design's two later prints read 1.029 and 1.045. Two standing conventions followed. First, under the *between-seat replication rule*, any single-seating value that would drive a decision requires confirmation on an independent re-seat. Second, we defined an advisory *seat gauge*, the broadband ratio of CFC-1000 to CFC-180 transmissibility. It reads 1.00 to 1.07 on most sessions, and we treat a session reading 1.15 or more as a re-seat candidate. The third print is the first replicate data collected after the gauge was defined (2026-09-12). Its one gross outlier (dran35, 10.1% above its design's median over the three prints) read 2.43, and its other eight sessions read 1.01 to 1.06. Over all 27 batch-3 sessions, the four reading 1.15 or more were each their design's highest value, and the gauge correlates with the deviation from the design median (Spearman $\rho = 0.52$, $p = 0.006$). Both conventions, with the session-by-session drift-watch record, live in the drop-test datasets on PR #86. The three-print comparison is archived with the manuscript data (`t3-prism-bo-round3-three-print.csv`, generated by `scripts/figures/round3_three_print_analysis.py`).

Objective provenance and pre-registered review. The objectives evolved through a measurability review (PR #97). Per-specimen crush energy in joules is not recoverable from this rig, because it has no falling-mass or displacement channel and the specimens stay elastic. The campaign responses are therefore the filtered peak-acceleration ratio and the mass-weighted rebound index, built on the square of a provisional rebound fraction, with quasi-static crush metrics deferred to a load-frame campaign. Batches 2 to 4 were recommended under the first power of the rebound fraction. We pre-registered an adversarial review of the objective choices, run with the Edison Scientific analysis agent (task `3e398131`, archived in the repository). It recommended accounting for specimen-level rather than per-drop noise, treating the rebound fraction as a candidate constraint pending a restrained-versus-unrestrained validation with synchronized video, and reserving print capacity for replicate specimens. The main text records how the campaign responded to these recommendations. We printed and tested the batch-3 plate three times, which measured the print-to-print scatter directly and produced the between-seat replication rule. The rebound fraction remains a candidate constraint, and the surrogate continues to ingest per-session standard errors, with the replication evidence carried as an interpretation bound rather than as an inflated noise term. We acted on the reserved-replicate recommendation with the replicate rank-stability study of Section S10, which re-printed three batch-4 designs nine times each to test whether the measured ordering survives print-to-print and seating variation. Its drop tests are not complete.

S8 Planned Quasi-Static Extension

The drop test loads the specimens in their elastic regime. The crush lasts 1 to 3 ms and the specimen recovers, so crush-energy metrics such as specific energy absorption (SEA) and compaction efficiency are not recoverable from these captures by any reprocessing. We plan a quasi-static compression campaign on a benchtop load frame, with a low-range load cell appropriate to the expected cell stiffness, to supply the complementary crush metrics

$$\mathrm{SEA} \;=\; \frac{1}{m}\int_0^{\delta_d} F(\delta)\,\mathrm{d}\delta, \qquad \eta_c \;=\; \frac{\int_0^{\delta_d} F(\delta)\,\mathrm{d}\delta}{F_{\max}\,\delta_d}, \tag{1}$$

with the densification displacement δ_d taken at the maximum of the energy-absorption efficiency curve and $F_{\max}$ the peak force within $[0, \delta_d]$. We follow the standard cellular-solid treatments [8–11] and mirror the quasi-static protocol of Pajunen et al. [1]. These metrics will be reported as a complementary characterization of selected designs rather than as objectives of the drop-tower campaign. Slip resistance and traction at the specimen-to-floor interface, governed by the tip geometry rather than the internal architecture, follow ISO 11334-4 [12] for walking-aid tips and are out of scope for the present axial-impact study.

S9 Printed-Mass Model

We projected the seed batch onto constant *solid* CAD mass ($m^\star = 30.95$ g), yet the specimens weigh 18.50 to 22.29 g. The gap arises because PLA struts print at partial infill (15% grid, two walls) while thin TPU cables print near solid, and the PLA-to-TPU volume split swings widely across the design box. A two-stage calibrated model closes this gap (PR #102, `bo/t3_prism_mass_model.py`). Analytic member volumes are corrected to rendered-solid grams (one factor per material). Rendered-solid grams are then mapped to weighed grams through a wall-plus-infill term in the as-printed strut diameter (fitted infill 23.6%, wall 1.18 mm, TPU factor 0.744). The model's residual standard deviation is 0.378 g. For context, the three available prints of design 08 have a sample standard

deviation of 0.457 g. Three prints of one geometry are insufficient to establish a general process repeatability limit. Run in reverse inside the projection, the model holds *printed* grams constant without running the slicer. For a target mass, the projection finds the uniform scale factor at which the predicted printed mass equals the target, and we print the design at that scale. Bisection is enough for this, because predicted mass rises monotonically with scale. Batches 3 and 4 used a target of 20.23 g, the mass of the printed reference specimen. Batch mass CV fell from 5.5% (tested seed specimens) and 8.7% (batch 2, still solid-mass-projected) to 1.7% (batch 3) and 2.3% (batch 4). The weighings sit at or below the model's estimate as a per-print flow offset (-0.52, -0.25, and -0.04 g for the three prints of batch 3 and -0.26 g for batch 4). In a first test of the model's infill terms, regressing the print-averaged mass residuals of the eighteen batch-3 specimens on the two commanded infills gives slopes indistinguishable from zero over the realized 13 to 34% spread. We will recalibrate the session term with the accumulated weighings before the next batch's projection. Two batch-2 recommendation sets exist in the record. The batch actually sliced and sent to the printer was generated before this correction, under the seed batch's solid-mass projection. A subsequently regenerated set applies the constant-printed-mass projection and seeded the batch-3 tooling. Both are archived with the manuscript data (`round2-as-printed-solid-mass-projection.csv`, `round3-candidate-constant-printed-mass.csv`).

S10 Replicate Rank-Stability Study

The replicate study re-prints three batch-4 designs that the lab chose on 2026-09-28, before any replicate was printed. Trial 37 (corny7) had the lowest transmissibility measured in the campaign. Trial 39 (corny8) is its near twin, differing mainly in TPU infill (34 against 21%). Trial 38 (corny2) comes from the tall, low-twist family at the far end of the batch-4 ranking. All three come from one batch, so one set of filament settings (PLA at 226 °C, TPU at 236 °C) reproduces every design's original process point on a single plate. Each plate holds three copies of each design in a 3×3 layout, and all three plates print the same G-code file. A specimen ID joins the plate number, the letters "sne", the last digit of the design's batch-4 ID, and the copy number, which gives the plate row (1 back, 2 middle, 3 front). For example, 1sne73 is the front-row corny7 copy on plate 1. The replicate print key in the repository maps every ID to its trial, plate position, and per-part infills.

The weighed masses of the 27 specimens are in the print log, which notes no defects on any of them. The photographs do show loose, light-colored strands on some TPU tendons of the plate-2 and plate-3 copies. The plate-1 copies weigh 0.5 to 0.9 g more than the batch-4 specimens of the same designs. The corny7 copies weigh 20.23 to 20.27 g against 19.62 g, the corny8 copies 20.11 to 20.30 g against 19.64 g, and the corny2 copies 20.89 to 21.08 g against 20.21 g. The plate-2 and plate-3 copies are up to 0.4 g lighter than those on plate 1. The replicate struts were printed in black PLA and the batch-4 struts in white PLA.

We drop-tested the nine plate-1 specimens on 2026-10-17, in one evening of 20-drop sessions under the standard protocol, and two faults affected the recording. From 17:18 to 18:17, an intermittent fault on the base-input channel (CH5), which also triggers the acquisition, set off the trigger away from the impact. Of the 70 captures in that hour, 56 are mistimed and contain no impact. The channel was healthy before and after that hour. Separately, every session on the six corny7 and corny8 copies showed a knocking top-vertex sensor. The knock is a spike of about 0.1 ms at impact onset, up to 3500 g, that reads the transmissibility high and fails the seat gauge of Section S7. In the three corny2 sessions the knock appeared on only two drops. On 2026-10-19 the input-channel instrumentation failed and drop testing stopped, with plates 2 and 3 printed and weighed but untested. Because the plate-1 record carries both faults and the instrument that

recorded it later failed, we withhold it until the repeats are complete on repaired instrumentation. The raw captures, the capture census, and the fault analysis are archived with the drop-test data.

S11 Metal-Analog Validation Plan

A planned validation campaign will rebuild a subset of the T_3-prism designs as metal analogs, using the same R, H, θ, and member-diameter parameterization as the printed PLA/TPU specimens. The analogs will use hollow aluminum rods for the compression members and stainless-steel threaded cables for the tension members. To probe transfer across material systems rather than only confirming the optimum, the selected designs will deliberately span the predicted performance range. Representatives the surrogate predicts to be worst-, mid-, and best-performing will each be built in both material systems and characterized under the identical drop protocol of Section S7. We treat this as a planned, exploratory comparison rather than a preregistered validation. The deformation modes, joint compliance, friction, pretension behavior, and strain-rate sensitivity all differ between the PLA/TPU and aluminum/steel systems, so agreement is not expected a priori. The protocol details that would let the comparison distinguish geometric rank transfer from material-system and mass confounds remain to be fixed and will be tabulated before any metal specimen is built. They are the aluminum alloy and temper, tube wall thickness, cable construction, joint hardware and torque, specimen and payload masses, replicate count, and tie-handling rules. The intended rank statistic is the Spearman rank-correlation coefficient ρ_s between the two material systems' orderings of the primary campaign metric (t_{180} under the identical drop protocol), reported with an exact permutation test. Mass-weighted rebound E_{reb} and, later, quasi-static specific energy absorption (SEA, Section S8) would give secondary rankings. Agreement between the two primary rankings would be preliminary evidence consistent with transfer beyond the as-printed polymers. Disagreement would not isolate which material or joint difference caused the change.

S12 Batch-3 Three-Print Repeatability Table

Table S6 tabulates the stabilized transmissibility of every batch-3 specimen over the three prints of the plate, the data behind the three-print figure and the replication findings of the main text (Section 4.2). The seat gauge and the drift flag used to screen the sessions are defined in Section S7.

S13 Surrogate Diagnostic Figures

Figures S2 and S3 are the surrogate diagnostics discussed in the main text (Section 4.3): the seed-state length-scale diagnostics of the sparse axis-aligned subspace model (SAASBO) and the leave-one-design-out cross-validation of the surrogate over the full 44-specimen record. The cross-validation runner, its fold-level records, the plotting script, and the reliability-ceiling analysis of the reprint pairs are in the project repository (`t3-prism-bo-round5-logocv.csv`, `scripts/figures/plot_logocv.py`, `scripts/figures/reprint_reliability_ceiling.py`).

S14 Simulation-Only Campaign

This section gives the simulator, search space, objective-pair selection, optimizer, baselines, and scoring behind the simulated campaign of the main text (Sections 3.5 and 4.4). Every number here and in Table S7 and Fig. S4 is recomputed from the per-seed run

Table S6: Batch-3 repeatability as stabilized t_{180} over three prints of the same nine designs, under the protocol of the main-text results tables (CFC-180, stabilized means). Specimens drran$N$, 2dran$N$, and dran3N are prints 1, 2, and 3. "Pred." is the posterior-mean t_{180} archived before any batch-3 specimen was tested. "Mean" averages the prints that pass the health screen, a seat gauge (the ratio of CFC-1000 to CFC-180 transmissibility) below 1.15 and no drift flag. § marks a failed seat gauge and † a transmissibility-drift flag. I_{reb} is listed for prints 1, 2, and 3.

Trial	N	Pred.	Print 1	Print 2	Print 3	Mean	I_{reb} (mg)
28	5	1.059	1.0373	1.0604	1.1679§	1.0489	5.6 / 5.5 / 11.6
29	4	1.032	1.0344	1.0433	1.0201	1.0326	18.6 / 27.8 / 52.5
30	9	1.058	1.0425	1.0451	1.0438	1.0438	6.3 / 52.1 / 10.1
31	3	1.043	1.0389	1.0533	1.0366	1.0429	4.0 / 19.8 / 10.7
32	7	1.020	1.2513§	1.0286	1.0448	1.0367	14.9 / 29.8 / 25.5
33	2	1.056	1.0345	1.0900§	1.0386	1.0366	32.0 / 4.6 / 4.7
34	6	1.065	1.0471	1.0859†	1.0353	1.0412	5.3 / 13.7 / 10.5
35	8	1.039	0.9804†	1.0109	0.9958	1.0033	29.2 / 22.5 / 15.4
36	1	1.051	1.0326	1.0791§	1.0209	1.0267	75.9 / 12.4 / 20.8

records by `scripts/figures/sim_campaign_tables.py`, which reads only the snapshot in `manuscript/data/sim/`. The snapshot is a verbatim copy of the simulation branch (`copilot/explore-simulations-for-tensegrity`, commit `94e53a8`) of the project repository (https://github.com/vertical-cloud-lab/tensegrity-optimization), which also holds the simulator and campaign code.

Simulator The drop tower is modeled in MuJoCo [13]. The falling base-plate assembly is a 2.0 kg carriage on a vertical slide joint that stands in for the guide rails, and it carries the base accelerometer site. The polyurethane impact surface is an explicit one-sided Hunt-Crossley contact on the carriage, $F = k\delta + \lambda\delta(-\dot{\delta})$ for penetration $\delta > 0$ and zero otherwise, with $k = 3.18 \times 10^5$ N/m and $\lambda = 5.40 \times 10^4$ N s/m^2. These two constants were calibrated once against the measured input peak and restitution of the reference prism (S0) and then held fixed for every design. The specimen is three rigid PLA strut capsules, which carry its whole printed mass, joined by nine TPU tendons, with its three bottom vertices ball-anchored to the carriage. The tendons are springs that carry tension only, with zero force below the slack length, a linear force above it, 2% prestrain, and a viscous damping ratio of 0.02. A 5 g mass at the top vertex stands for the accelerometer. Each drop starts at the measured campaign impact velocity of 5.30 m/s (the 60 in drop) and runs for 40 ms at a 20 µs time step. The simulator reports t_{180} as the ratio of the CFC-180-filtered peak accelerations at the top vertex and the base. The model omits strut flexure, mount compliance, tendon hysteresis, and drift in the impact surface. Because the struts are rigid and cannot resonate, simulated t_{180} stays below about 1, whereas most printed specimens measured above 1, so the simulator benchmarks the optimizer and is not a substitute for the drop tower.

Search space and constraints The search varies four scale-free shape ratios: $H/R \in [1.5, 4.4]$, $H/d_s \in [5.0, 18.3]$, $d_t/d_s \in [0.25, 0.92]$, and twist $\theta \in [40, 80]^\circ$. These are the extremes the five-variable box of the physical campaign can express. For each design, the printed-mass model (Section S9) solves in closed form for the overall scale at which the design prints at 20.23 g, the weighed mass of S0, so mass never enters the search. The printed tendon diameter must be at least 3.0 mm and the envelope volume $\pi R^2 H$ at most 250 cm^3, the two printability screens of Section S2, given

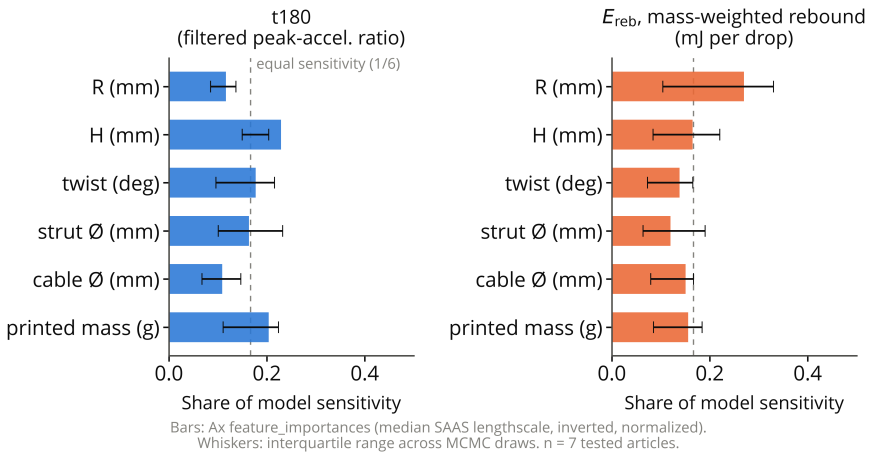


Figure S2: Seed-state SAASBO length-scale diagnostics per objective for the five design variables plus printed mass. Bars are median normalized inverse length scales over MCMC draws, and whiskers are interquartile ranges. The horizontal axis, labeled "share of model sensitivity" in the plot, is this normalized inverse-length-scale diagnostic. No input separates decisively from the equal-share line (dashed) at seven mapped specimens, and these prior-sensitive model diagnostics are not evidence of variable relevance.

to the optimizer as outcome constraints.

Choice of objective pair Before the campaign, one scrambled Sobol sweep of 16,384 designs (seed 20260825, distinct from every campaign seed; 9,738 printable) priced six candidate objective pairs at once, since one simulation returns every candidate observable. For each pair, 2,000 random batches of 45 printable designs measured the fraction of the sweep's hypervolume that an uninformed batch collects. The rule, fixed before any campaign run, excluded the two existing pairs as controls, required the front to span a meaningful part of both objectives, and took the lowest such fraction. That selected t_{180} with envelope volume (80.0% collected by a random batch), against 84.6% for the t_{180} and peak-tendon-strain control. The selection favors pairs on which a model can help, which is why the main text reports the control pair alongside it.

Optimizer and baselines Each of ten seeds (0 to 9) ran Ax 0.5.0, starting from a nine-design scrambled Sobol batch seeded by the repeat's own seed. Four batches of nine followed from Ax's modular BoTorch model, a single-task GP with constrained noisy expected hypervolume improvement (qNEHVI) [14], with the acquisition optimized from 8 restarts and 128 raw samples. No two seeds share a first batch. The baselines are uniform random sampling, scrambled Sobol, scrambled Latin hypercube sampling [15], and compass search [16], each at the same 45-design budget and the same seeds. In the filtered mode reported in the main text, every baseline candidate passes the closed-form printability check before it is simulated, and compass search skips unprintable probes without charge. No baseline therefore spends budget on a design that cannot be printed. In the plain mode, unprintable draws are simulated, charged, and excluded from the front, as in the earlier strain-pair runs that serve as the control.

Scoring Both objectives are minimized. Each pair has one reference point, 1.05 times the componentwise worst value over the printable sweep: (1.039, 262.4 cm^3) for the envelope pair and (1.039,

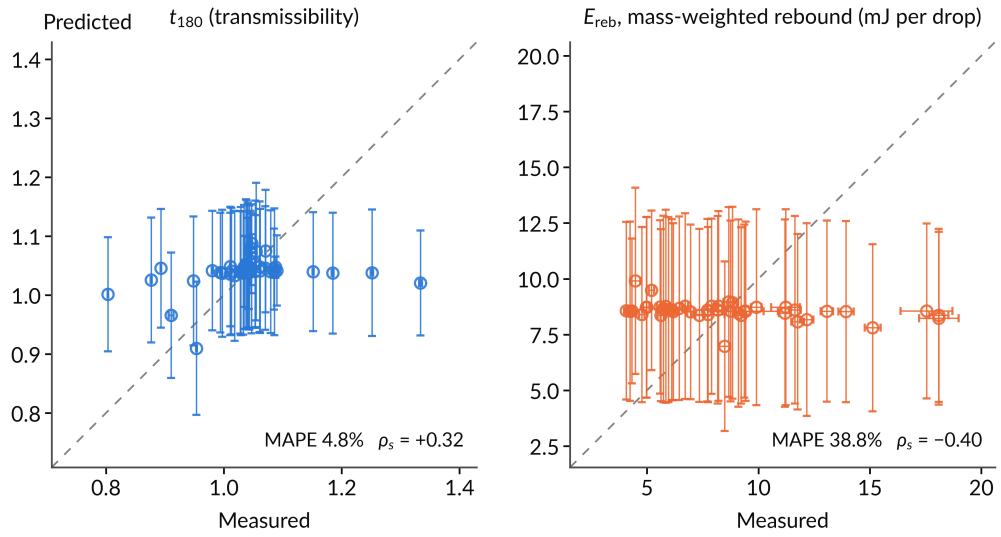


Figure S3: Leave-one-design-out cross-validation of the SAASBO surrogate over the full 44-specimen record (35 design folds, one NUTS refit per fold at 256 samples after 512 warmup steps, the budget of the generation fits). Each batch-3 reprint pair is held out together so neither print leaks into the other's prediction. Vertical bars are one posterior standard deviation of the held-out prediction, horizontal bars are the ingested standard error, and the dashed line is parity. Held-out t_{180} predictions stay close to the pooled mean (MAPE 4.8%, Spearman $\rho_s = +0.32$, two-sided permutation $p = 0.036$), and the ordering of E_{reb}, the first-power rebound objective these fits used, runs against the measurements (MAPE 38.8%, $\rho_s = -0.40$, $p = 0.008$). Design-level held-out skill is weak on transmissibility and absent on E_{reb}. The folds, held-out predictions, and diagnostics are archived as `t3-prism-bo-round5-logocv.csv` and its `-diagnostics.json`, and the earlier run at 64 samples after 128 warmup steps under the suffix `-nuts64` (t_{180} $\rho_s = +0.08$, E_{reb} $\rho_s = -0.38$).

0.247) for the strain pair. A run's hypervolume is that of its feasible evaluations against this point. The denominator is a best-known reference hypervolume, the hypervolume of the union of the printable sweep, a Nelder-Mead refinement of the sweep's front, and every feasible evaluation of the compared runs. For the envelope pair this is 87.09 cm^3, 1.35% above the refined front alone (85.93 cm^3). BO's own evaluations account for 1.14 of the 1.16 cm^3 increase, so scoring against the refined front alone would raise BO to 94.2% and the baselines to 77.5 to 80.2%. Adding the plain-mode baseline evaluations to the union changes the denominator by less than 0.01%. For the strain pair the best-known reference is 0.1% above the refined front. The main test is the one-sided exact Mann-Whitney test on the ten final hypervolumes per method, whose floor at ten against ten is $1/\binom{20}{10} = 5.4 \times 10^{-6}$. We also report the paired exact Wilcoxon signed-rank test, with runs paired by seed index, whose floor at ten pairs is 9.8×10^{-4}. Every envelope-pair comparison, in both baseline modes, reaches both floors. On the strain pair, the Wilcoxon p values run from 0.014 to 0.12 and BO wins 5 to 9 of the ten pairs.

Audit The script recomputes each run's hypervolume from its evaluations and checks it against the running value stored in the record (largest relative difference 2×10^{-6}, from rounding in the CSV files). It also checks the strain-pair rescoring against the committed per-seed values, that every run holds exactly 45 evaluations, and that no two BO seeds drew the same first batch. An independent review of the campaign code and records (Edison analysis task `53cfc937`, archived

with the snapshot) cross-checked the hypervolume routine against BoTorch's exact implementation and proposed the best-known denominator and the exact tests used here.

Table S7: Simulation-only campaign, t_{180} with envelope volume. HV is the final hypervolume after 45 designs as a percentage of the best-known reference (mean ± sample standard deviation), and Range spans the ten seeds. Best t_{180} and Min. envelope are means over seeds of each run's best values. p is the one-sided exact Mann-Whitney test against BO, and Wins counts the seeds on which BO finished ahead. In the Mode column, filtered baselines skip unprintable candidates without simulating them, and plain baselines simulate and charge them.

Method	Mode	HV (%)	Range (%)	Best t_{180}	Min. envelope (cm^3)	p	Wins
BO (qNEHVI)		93.0 ± 2.2	90.0 to 96.1	0.691	11.0		
Random	filtered	79.1 ± 2.7	75.8 to 84.2	0.704	23.7	5.4×10^{-6}	10/10
	plain	74.1 ± 5.8	64.6 to 82.0	0.715	29.4	5.4×10^{-6}	10/10
Sobol	filtered	77.3 ± 4.2	69.9 to 82.2	0.703	20.8	5.4×10^{-6}	10/10
	plain	73.2 ± 3.4	65.8 to 76.4	0.713	26.2	5.4×10^{-6}	10/10
Latin hypercube	filtered	76.5 ± 3.4	69.9 to 81.3	0.699	25.2	5.4×10^{-6}	10/10
	plain	72.0 ± 3.4	65.3 to 75.7	0.712	29.2	5.4×10^{-6}	10/10
Compass search	filtered	78.1 ± 5.3	66.5 to 83.2	0.717	12.3	5.4×10^{-6}	10/10
	plain	71.5 ± 10.8	52.0 to 87.3	0.739	23.7	5.4×10^{-6}	10/10

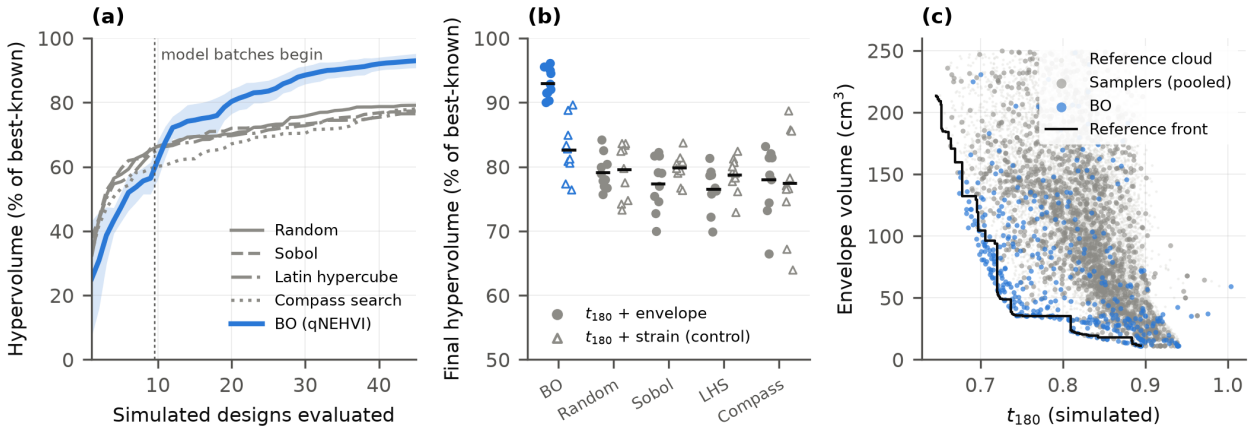


Figure S4: Simulation-only campaign. (a) Running hypervolume of the t_{180} and envelope-volume pair, scaled to the best-known reference and averaged over ten seeds (band: ± one standard deviation for BO), with filtered baselines. BO's first nine designs are an unfiltered Sobol batch, which is why it starts below the baselines. (b) Final hypervolume of every seed for both objective pairs (bar: mean). On the envelope pair every BO seed finishes above every baseline seed; on the strain control the distributions overlap. (c) Feasible evaluations of all seeds pooled in objective space, with the printable reference sweep in light gray and the refined reference front as a staircase. BO's evaluations follow the front along its length, and some lie beyond it, which is why the best-known reference includes them.

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